mentally operate within a knowledge-dependent collaborative framework between humans and AI.

AI Scientist represents a research paradigm shift where AI assumes the role of an autonomous scientist capable of conducting independent scientific research. As illustrated in Figure 2, while scientific tools operate under human supervision, receiving data as input and producing predictions as output, AI Scientist goes a step further by demonstrating autonomous scientific reasoning capabilities. It accepts research questions as input and engages in iterative, self-directed interactions with scientific tools to generate comprehensive solutions. Unlike scientific tools that function as sophisticated instruments awaiting human guidance, AI Scientist exhibits genuine scientific agency, conducting end-to-end scientific investigations from question formulation to solution discovery (Yamada et al., 2025).

2.2. Conceptualized Framework

> ***Our Definition:*** *An AI Scientist is an advanced end-to-end system capable of independently formulating scientific ideas and executing the requisite verification and falsification procedures.*

We define an AI Scientist, denoted as $\mathcal{S}_{AI}$, as a fully autonomous scientific intelligence capable of independently performing diverse scientific research tasks. Different from a general scientific tool, it must possess dual capacities, including idea generation and experimental execution. A complete scientific research task typically originates from an initial scientific question $\mathcal{Q}_{init}$ and leverages existing domain knowledge $\mathcal{K}_{domain}$. An AI Scientist, denoted $\mathcal{S}_{AI}$, operates within the scope of human ethical constraints $\mathcal{R}_{human}$ and resource constraints $\mathcal{B}_{res}$ to conduct this task. The primary output is the generation of novel scientific knowledge $\mathcal{K}_{new}$ and associated verifiable artifacts $\mathcal{A}_{sci}$. The process through which an AI Scientist aims to achieve the optimal output from a scientific research task can be formally represented as:

$$(\mathcal{K}_{new}, \mathcal{A}_{sci}) \leftarrow \max\{\mathcal{S}_{AI}(\mathcal{Q}_{init}, \mathcal{K}_{domain}, \mathcal{R}_{human} | \theta_{AI}, \mathcal{B}_{res})\} \tag{1}$$

3. Arguments for Implementation Capability

We argue that the fundamental bottleneck limiting AI Scientists lies not in their idea generation capabilities, but in their capacity to execute rigorous implementation procedures required for reliable scientific research. To support this position, we present three lines of evidence: systematic analysis of research trends in the AI Scientist literature (Section 3.1), comprehensive benchmark analysis across multiple evaluation frameworks (Section 3.2), and systematic peer review assessment using LLM-as-a-Judge methodology (Section 3.3).

3.1. Research Trend of AI Scientist

Our statistical analysis of AI Scientist papers on arXiv up to May 23, 2025 (see Appendix B for details), reveals key trends illustrated in Figure 3. The lower panel of the figure shows that while the total number of publications is growing, studies focusing on idea generation without concrete implementation details consistently outnumber those incorporating such implementations. Despite this disparity in publication

Table 1: State-of-the-art (SoTA) LLMs show relatively low accuracy on code implementation on different tasks. The listed benchmarks are collected from diverse domains. The table below details their tasks, domains, scale, methods, and performance.

Benchmark	Task Description	Domains	Scale	LLM	Acc. Performance
MLE-Bench(Chan et al., 2024)	AI Training task	Applied ML	75	OpenAI o1-preview	16.90%
PaperBench (Starace et al., 2025)	ICML paper Replicating	NLP, CV, ML	8,316	OpenAI o1-high	26.00%
SciReplicate-Bench (Xiang et al., 2025)	Code Generation	NLP	100	Claude-Sonnet-3.7	39.00%
CORE-Bench (Siegel et al., 2024)	Scientific Paper reproduction	Computer Science, Social Science, and Medicine	270	OpenAI GPT-4o	55.56%
ML-Dev-Bench (Padigela et al., 2025)	AI training task	ML	30	Claude-Sonnet-3.5	50.00%

numbers, the upper panel indicates a crucial counterpoint: papers that include substantive implementation details achieve a significantly higher average number of citations. This signals strong community valuation for executable advancements and underscores the importance of addressing the implementation gap. This then raises a critical question: if implementation-focused research garners higher impact, why does its volume remain markedly lower? This disparity strongly implies that the path of implementation is fraught with substantial challenges.

3.2. Quantitative Analysis

Empirical Evidence of Implementation Gap. Advanced LLMs achieve near-saturated performance on simple code generation benchmarks like HumanEval (Chen et al., 2021, Liu et al., 2023, Yang et al., 2025a). For example, o3 exhibits excellent problem-solving capabilities in the 99.8th percentile of human performance on algorithmic competition platforms like Codeforces. However, the performance of SoTA LLMs **drops dramatically when it comes to real-world research scenarios**. As depicted in Table 1, we compare existing benchmarks for evaluating LLM agents' abilities to perform challenging machine learning research tasks, including MLE-Bench (Chan et al., 2024) (solving Kaggle machine learning tasks, evaluated by medal rates), PaperBench (Starace et al., 2025) (replicating ICML research papers from scratch, assessed by replication scores), SciReplicate-Bench (Xiang et al., 2025) (generating executable code from NLP algorithm descriptions, measured by execution accuracy), CORE-Bench (Siegel et al., 2024) (reproducing computational results from scientific papers, determined by accuracy), and ML-Dev-Bench (Padigela et al., 2025) (completing diverse ML development workflow tasks, assessed by success rates). Each takes a different approach to measuring how well AI systems can automate aspects of ML research. These evaluations consistently demonstrate that LLMs face difficulty in translating conceptual understanding or initial plans into

Type	Total citations	Avg. citations
Without Implementation	216	10.3
With Implementation	**325**	**25.0**

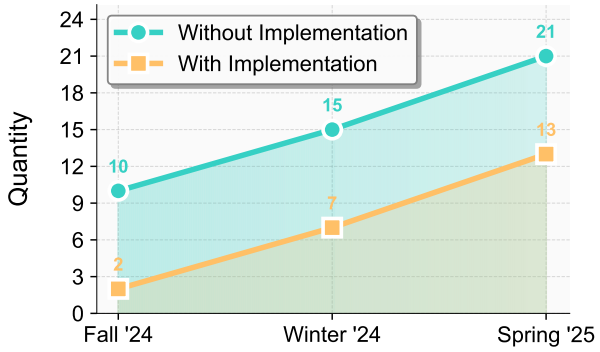


Figure 3: Analysis of AI Scientist publications on arXiv. The upper panel displays the average number of citations up to now, categorized by containing implementation details. The lower panel shows the growth in the total number of these papers with the same categorization.

verifiably correct and operational code. **This "implementation gap" fundamentally limits AI Scientist's verification capabilities.**

Beyond Code Generation. The complexity of real-world research implementation processes extends far beyond simple code generation tasks, often requiring sustained reasoning and multi-step problem-solving. However, current LLMs exhibit relatively weak performance on such complex challenges. LiveCodeBench (LCB) (Jain et al., 2024), a more complex evaluation benchmark than Humaneval (Chen et al., 2021) that collects problems from periodic contests on LeetCode, AtCoder, and Codeforces platforms, evaluates Code LLMs across diverse code-related scenarios, including code generation, execution, self-repair, and output prediction. o4-mini achieves SoTA performance on the code generation subtask with **only 52.1**% **pass@1 score**. **This poor performance on complex coding tasks reveals that AI scientists lack the implementation ability to handle sophisticated code-based research scenarios.**

Implementation and verification. We observe that the verification bottleneck emerges across multiple stages of the research process. SciReplicate-Bench (Xiang et al., 2025), which tasks LLM agents with generating Python code to reproduce algorithms from NLP research papers, reveals that despite agents demonstrating an understanding of algorithmic logic (evidenced by high reasoning graph accuracy), they struggle with code execution. The best agent achieved only **39**% execution accuracy, indicating its generated code passed functional test cases for just **39**% of the tasks, highlighting **a failure to ensure implementation correctness and runtime behavior.** Similarly, PaperBench (Starace et al., 2025) requires LLM agents to replicate entire machine-learning papers from scratch by developing codebases and running experiments. While agents can generate code components (e.g., o1-High achieving **43.4**% success on weighted "Code-Development" sub-tasks), their performance on subsequent verification stages is poor. On rubric-defined leaf nodes for "Execution" (successfully running the code) and "Result Match" (quantitatively matching the paper's reported results), Claude 3.5 Sonnet scored only **1.8**% and **0.7**% respectively. This poor performance indicates a breakdown in ensuring the developed solution operates correctly and produces the intended outcomes.

Discussion. The verification challenge extends beyond initial code implementation to debugging, iterative refinement, and validation of experimental outcomes. Evidence from MLE-Bench and ML-Dev-Bench (Chan et al., 2024, Padigela et al., 2025) shows that LLM agents frequently fail to debug their code or produce valid submissions, with **20%** of o1 preview runs on MLE Bench failing this step, and struggle to optimize model performance. Debugging, an explicit verification procedure, also indicates persistent agent failures that highlight the verification bottleneck (Chan et al., 2024). The incapacity to iteratively refine solutions towards better performance, illustrated in ML-Dev-Bench where all tested agents scored **0%** on "Model Performance" tasks, further signifies deficiencies in robust verification loops essential for scientific advancement (Padigela et al., 2025). Furthermore, CORE-Bench, which requires agents to reproduce results and then answer questions based on these outputs, assesses the verification of entire computational experiments. This process, involving multiple stages of reproduction and reasoning, presents significant challenges. For instance, the imperfect success rates (e.g., CORE-Agent with GPT-4o achieved **55.56%** on CORE-Bench Medium) highlight the difficulties in this complex verification process (Siegel et al., 2024). These difficulties across verification tasks suggest that enhancing AI Scientists' systematic verification capability is crucial for their maturation into ideal AI Scientists. Current LLMs, while proficient in content generation, fail to rigorously validate their outputs against explicit criteria, a foundational component of scientific practice.